

q-Binomial Richardson Acceleration of Geometric Sinc Products

Bernoulli–Bell tail expansions, moment-matched Rvachev splines,
and interior inverse-Fabius asymptotics

Frontier research report for the PROVEIT Fabius Function project

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Abstract

The Fourier transform of Rvachev’s compactly supported up-function is the geometric sinc product

$$P(\xi) = \prod_{\nu \geq 1} \text{sinc}(\xi/2^\nu).$$

The repository corpus develops its large-frequency decay, its connection with the Fabius function and the signed Thue–Morse sequence, dyadic exact values, small-argument and inverse asymptotics, and several Richardson-type limits. This report studies a different asymptotic variable: the truncation level in the finite product $P_n(\xi) = \prod_{\nu=1}^n \text{sinc}(\xi/2^\nu)$.

For a general geometric base $b > 1$, with $q = b^{-2}$, we prove an exact analytic factorization

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \exp\left(\sum_{k \geq 1} c_k(b) \xi^{2k} q^{kn}\right) = \sum_{r \geq 0} \alpha_r(b) \xi^{2r} q^{rn}.$$

The cumulants $c_k(b)$ are explicit Bernoulli-number combinations and the coefficients $\alpha_r(b)$ are complete Bell polynomials. Lagrange evaluation at the geometric nodes $1, q, \dots, q^m$ then gives a universal extrapolant

$$\mathcal{R}_{m,n} P_b = \sum_{j=0}^m w_j^{(m)} P_{b,n+j}, \quad w_j^{(m)} = \frac{(-1)^{m-j} q^{(m-j)(m-j+1)/2}}{(q; q)_j (q; q)_{m-j}}.$$

Its complete error series has the closed q-binomial form

$$\frac{\mathcal{R}_{m,n} P_b}{P_b} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) \xi^{2r} q^{rn}.$$

Consequences include an exact sign law, non-asymptotic divided-difference bounds, a rigorous a posteriori error certificate, and a uniformly bounded stability constant $\prod_{r=1}^m (1 + q^r)/(1 - q^r)$. For $q = 1/4$ this constant stays below 1.970, even as the order tends to infinity.

After inverse Fourier transformation, the same filter combines consecutive finite box splines. It matches the first $2m$ moments of the limiting up-density *exactly at every truncation level*, and converges in every C^s -norm at rate $q^{\binom{m+1}{2}n}$. The first error profile is an explicit even derivative of the up-function. Translating to the Fabius function gives the same accelerated approximation, while inversion on compact interior intervals yields an explicit first correction for the inverse Fabius function. These statements are proved here and checked symbolically and numerically by the accompanying Python program. They are described as *repository-new*: the claim is that they do not occur in the audited `*.tex` corpus under the stated repository path, not that a priority search over the entire literature has been completed.

Status of results. Every result labelled theorem, proposition, lemma, or corollary below has a proof in this report. The numerical tables are reproducible from `verify_frontier_results.py`. No claim in this report has yet been formalized in Lean.

Contents

1	Audit scope and the non-duplication boundary	3
1.1	How the corpus was used	3
1.2	What is new in this report	3
2	Geometric sinc products and finite Rvachev splines	4
2.1	The finite Thue–Morse derivative comb	4
3	The exact Bernoulli–Bell tail expansion	5
3.1	A universal tail function	5
3.2	Bernoulli cumulants	5
3.3	Bell coefficients and their recurrence	6
4	q-Lagrange Richardson extrapolation	7
4.1	The filter polynomial	7
4.2	A Romberg-style triangular algorithm	7
4.3	The exact q-moment identity	8
4.4	The complete error series	8
4.5	Divided differences and non-asymptotic sign bounds	9
4.6	Uniform stability of arbitrarily high order	9
4.7	A rigorous a posteriori certificate	10
5	Physical-space consequences for Rvachev splines	10
5.1	All-order inverse-differential expansion	11
5.2	Superconvergence of the extrapolated spline	11
5.3	Exact moment matching	12
6	Fabius and inverse-Fabius consequences	12
6.1	Relation to Lambert– W endpoint asymptotics	13
7	Numerical verification	13
7.1	Absolute errors	13
7.2	Leading constants	14
7.3	A posteriori estimates	14
8	A universal geometric-product theorem	14
9	Lean formalization roadmap	15
10	Further frontier problems	16
10.1	A matched Lambert–Richardson endpoint theory	16
10.2	Optimal order in finite precision	16
10.3	Positivity restoration	16
10.4	Dyadic rational evaluation from accelerated splines	16
10.5	Complex domains and pole-aware extrapolation	17
10.6	Other refinable and compactly supported functions	17

11 Conclusion	17
A Exact coefficients and filters at $b = 2$	17
B Corpus audit map	18
C Reproducibility notes	19

1 Audit scope and the non-duplication boundary

1.1 How the corpus was used

The source audit covered every LaTeX family in the repository documentation tree [1], including the current expository volumes, the formalization walkthrough, the non-elementarity article, the mathematical glossary, the consolidated frontier volume, the Thue–Morse formula atlas, all waves of the Fourier-decay discussion, the paper-source subdirectories, and the historical archive. The archive index was used to track renamed and byte-identical historical variants, so that an old statement was not mistaken for a new one merely because it appeared under a superseded filename. A source map is given in [section B](#).

The audit revealed four especially important boundaries.

- B1.** The main exposition and its supplements already contain the normalized infinite sinc product, the box-convolution construction of up, its relation to the Fabius function, dyadic identities, q-Pochhammer and q-binomial formulae, and the signed Thue–Morse derivative comb.
- B2.** The asymptotic documents already contain a corrected Lambert– W endpoint scale, a periodic dyadic phase, and all-order small-argument expansions for the Fabius function and its inverse. Those results concern $x \downarrow 0$, not finite-product truncation $n \rightarrow \infty$.
- B3.** The Fourier-decay waves study $|P(\xi)|$ as $|\xi| \rightarrow \infty$, including dyadic oscillation and exceptional frequencies. They do not develop the exact analytic expansion of $P_{b,n}/P_b$ in the small geometric variable q^n , nor the extrapolation filter derived below.
- B4.** The frontier volume already has q-Richardson ideas for discrete Fabius spline rows and degree-dependent approximants. We therefore do not present a generic Richardson recipe as new. The new object here is the *Fourier-tail truncation sequence* $P_{b,n}$, for which the error coefficients, all q-moments, stability constant, moment matching, and inverse-function consequence can all be written exactly.

The external search performed for this report also found general literature on Richardson extrapolation and on powers of sinc expressed through Bell polynomials, but no source matching the combined construction in [sections 3 to 5](#). This is supporting evidence only, not an exhaustive priority claim.

1.2 What is new in this report

The principal contributions are the following.

- N1.** A one-variable analytic tail function Φ_b whose Taylor cumulants are rational Bernoulli combinations and whose Taylor coefficients are complete Bell polynomials.
- N2.** A q-Lagrange operator $W_m(T_q) = \prod_{r=1}^m (T_q - q^r)/(1 - q^r)$ that annihilates the first m truncation modes exactly.

- N3.** An all-order q-binomial error identity, not merely a leading-order Richardson estimate.
- N4.** A closed stability constant with a finite infinite-order limit, and a successive-level error certificate requiring no knowledge of the infinite product.
- N5.** Exact matching of the first $2m$ moments of the limiting Rvachev density by a signed combination of finite box splines, for every n .
- N6.** A C^∞ superconvergence theorem with an explicit first derivative profile, followed by an interior inverse-Fabius perturbation formula.
- N7.** A kernel-independent extension showing that the same q-filter accelerates any geometric product whose elementary factor has an even logarithmic Taylor series.

2 Geometric sinc products and finite Rvachev splines

Fix $b > 1$ and set

$$q = b^{-2} \in (0, 1), \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Define

$$P_b(\xi) = \prod_{\nu=1}^{\infty} \text{sinc}(\xi/b^\nu), \quad P_{b,n}(\xi) = \prod_{\nu=1}^n \text{sinc}(\xi/b^\nu). \quad (1)$$

The convergence is locally uniform in $\mathbb{C}$, since $1 - \text{sinc}(\xi/b^\nu) = O(b^{-2\nu})$ on compact sets.

Let

$$h_{b,\nu}(x) = \frac{b^\nu}{2} \mathbf{1}_{[-b^{-\nu}, b^{-\nu}]}(x), \quad u_{b,n} = h_{b,1} * \cdots * h_{b,n}. \quad (2)$$

With the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(\xi) e^{i\xi x} d\xi,$$

we have $\widehat{h}_{b,\nu}(\xi) = \text{sinc}(\xi/b^\nu)$, hence

$$\widehat{u_{b,n}} = P_{b,n}. \quad (3)$$

The infinite convolution $u_b = *_{\nu \geq 1} h_{b,\nu}$ is a probability density supported on $[-1/(b-1), 1/(b-1)]$, and $\widehat{u}_b = P_b$. The product decays faster than every power, so $u_b \in C_c^\infty(\mathbb{R})$. At $b = 2$, u_2 is the normalized Rvachev up-function.

2.1 The finite Thue–Morse derivative comb

The bridge to the signed Thue–Morse sequence is already part of the repository background, but the exact normalization is useful for what follows. Write $t_k = (-1)^{s_2(k)}$, where $s_2(k)$ is the binary digit sum.

Proposition 2.1 (distributional Thue–Morse comb). *For $n \geq 1$,*

$$D^n u_{2,n} = 2^{n(n-1)/2} \sum_{k=0}^{2^n-1} t_k \delta_{-1+(2k+1)2^{-n}}. \quad (4)$$

Proof. For $a > 0$,

$$D \left(\frac{1}{2a} \mathbf{1}_{[-a,a]} \right) = \frac{1}{2a} (\delta_{-a} - \delta_a).$$

Differentiate every factor in (2). Choosing the right endpoint in the j -th factor contributes one minus sign. If the choices are encoded by the binary digits of k , the coefficient is $(-1)^{s_2(k)}$, while the location is

$$-\sum_{j=1}^n 2^{-j} + 2 \sum_{j=1}^n \varepsilon_j 2^{-j} = -1 + 2^{-n} + (2^{1-n})k.$$

Finally $\prod_{j=1}^n (2 \cdot 2^{-j})^{-1} = 2^{n(n-1)/2}$. $\square$

Thus every finite product in (1) is simultaneously a Fourier approximation and an integrated Thue–Morse object. The acceleration below acts on both descriptions at once.

3 The exact Bernoulli–Bell tail expansion

3.1 A universal tail function

The key observation is that the two variables n and ξ enter the omitted tail only through $q^n \xi^2$.

Definition 3.1. For z near zero, define

$$\Phi_b(z) = \prod_{\ell=1}^{\infty} \frac{1}{\text{sinc}(b^{-\ell} \sqrt{z})}. \quad (5)$$

The expression is independent of the sign of $\sqrt{z}$, and hence defines an ordinary analytic function of z .

Theorem 3.2 (exact tail factorization). *The function Φ_b is analytic in $|z| < \pi^2 b^2$, meromorphic in $\mathbb{C}$, and has its nearest singularity to the origin at $z = \pi^2 b^2$. For every $n \geq 0$,*

$$P_{b,n}(\xi) = P_b(\xi) \Phi_b(q^n \xi^2). \quad (6)$$

The identity is entire in ξ ; at common zeros it is interpreted by analytic continuation. Moreover,

$$\Phi_b(z) = \frac{\Phi_b(qz)}{\text{sinc}(b^{-1} \sqrt{z})}. \quad (7)$$

Proof. Cancel the first n factors in $P_b/P_{b,n}$ and write $\nu = n + \ell$:

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \prod_{\ell=1}^{\infty} \frac{1}{\text{sinc}(\xi/b^{n+\ell})} = \Phi_b(q^n \xi^2).$$

The zeros of $\text{sinc}(b^{-\ell} \sqrt{z})$ occur at $z = \pi^2 k^2 b^{2\ell}$, $k \in \mathbb{Z} \setminus \{0\}$. The nearest is $\pi^2 b^2$. Splitting off the $\ell = 1$ factor and reindexing proves (7). $\square$

3.2 Bernoulli cumulants

The classical expansion

$$\log \text{sinc } w = - \sum_{k=1}^{\infty} \frac{\zeta(2k)}{k \pi^{2k}} w^{2k}, \quad |w| < \pi, \quad (8)$$

turns the geometric tail into an explicitly summable cumulant series.

Theorem 3.3 (Bernoulli cumulants). *For $|z| < \pi^2 b^2$,*

$$\Phi_b(z) = \exp\left(\sum_{k=1}^{\infty} c_k(b) z^k\right), \quad (9)$$

where

$$c_k(b) = \frac{\zeta(2k)}{k\pi^{2k}(b^{2k} - 1)} \quad (10)$$

$$= \frac{(-1)^{k+1} 2^{2k-1} B_{2k}}{k(2k)!(b^{2k} - 1)}. \quad (11)$$

In particular $c_k(b) > 0$ for every $k \geq 1$.

Proof. Insert (8) into (5), interchange absolutely convergent sums, and use

$$\sum_{\ell=1}^{\infty} b^{-2k\ell} = \frac{1}{b^{2k} - 1}.$$

The Bernoulli form follows from Euler's formula for $\zeta(2k)$. Positivity is clear from (10). $\square$

Combining theorems 3.2 and 3.3 gives the exact truncation expansion

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \exp\left(\sum_{k \geq 1} c_k(b) \xi^{2k} q^{kn}\right). \quad (12)$$

This is an equality, not a merely formal asymptotic series, whenever $|\xi| < \pi b^{n+1}$.

3.3 Bell coefficients and their recurrence

Write

$$\Phi_b(z) = \sum_{r=0}^{\infty} \alpha_r(b) z^r, \quad \alpha_0(b) = 1. \quad (13)$$

Proposition 3.4 (complete Bell-polynomial formula). *Let Y_r denote the complete exponential Bell polynomial. Then*

$$\alpha_r(b) = \frac{1}{r!} Y_r(1!c_1(b), 2!c_2(b), \dots, r!c_r(b)), \quad (14)$$

and equivalently

$$r\alpha_r(b) = \sum_{k=1}^r k c_k(b) \alpha_{r-k}(b). \quad (15)$$

Every $\alpha_r(b)$ is strictly positive.

Proof. Equation (14) is the defining exponential generating identity for complete Bell polynomials. Differentiating (9) and comparing coefficients gives (15). Positivity follows inductively from $c_k(b) > 0$. $\square$

For the Rvachev base $b = 2$, the first cumulants are

$$\begin{aligned} c_1 &= \frac{1}{18}, & c_2 &= \frac{1}{2700}, & c_3 &= \frac{1}{178605}, \\ c_4 &= \frac{1}{9639000}, & c_5 &= \frac{1}{478533825}, & c_6 &= \frac{691}{15688261338750}. \end{aligned}$$

The corresponding Bell coefficients are

$$\alpha_0 = 1, \quad \alpha_1 = \frac{1}{18}, \quad \alpha_2 = \frac{31}{16200}, \quad (16)$$

$$\alpha_3 = \frac{2347}{42865200}, \quad \alpha_4 = \frac{1904369}{1311675120000}, \quad \alpha_5 = \frac{3309760579}{88561680752160000}. \quad (17)$$

The exact rationals are useful because no numerical fitting is needed at any order.

4 q-Lagrange Richardson extrapolation

4.1 The filter polynomial

Let T_q be the dilation operator $(T_q f)(z) = f(qz)$. Define

$$W_m(t) = \prod_{r=1}^m \frac{t - q^r}{1 - q^r}, \quad W_0(t) = 1. \quad (18)$$

Thus $W_m(1) = 1$ and $W_m(q^r) = 0$ for $1 \leq r \leq m$. Expanding $W_m(t) = \sum_{j=0}^m w_j^{(m)} t^j$ gives the following exact weights.

Theorem 4.1 (geometric Lagrange weights). *For $0 \leq j \leq m$,*

$$w_j^{(m)} = \prod_{\substack{0 \leq \ell \leq m \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} = \frac{(-1)^{m-j} q^{(m-j)(m-j+1)/2}}{(q; q)_j (q; q)_{m-j}}, \quad (19)$$

where $(q; q)_r = \prod_{s=1}^r (1 - q^s)$. These are precisely the Lagrange weights that recover the value at zero from data at $1, q, \dots, q^m$.

Proof. The first expression is the Lagrange cardinal basis evaluated at zero. Separating factors with $\ell < j$ and $\ell > j$, then collecting powers of q , gives the second expression. Alternatively, the q-binomial theorem expands (18) directly. $\square$

Definition 4.2 (accelerated finite product). For $m, n \geq 0$, define

$$\mathcal{R}_{m,n} P_b(\xi) = \sum_{j=0}^m w_j^{(m)} P_{b,n+j}(\xi). \quad (20)$$

With $z = q^n \xi^2$, the tail factorization gives the compact operator identity

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} = (W_m(T_q) \Phi_b)(z). \quad (21)$$

The right side is well defined as an analytic continuation even at common zeros.

4.2 A Romberg-style triangular algorithm

The factorization of W_m gives a recursion that avoids an explicit weighted sum.

Algorithm 4.3 (q-Richardson table). Start with $A_{0,n} = P_{b,n}(\xi)$. For $m \geq 1$, form

$$A_{m,n} = \frac{A_{m-1,n+1} - q^m A_{m-1,n}}{1 - q^m}. \quad (22)$$

Then $A_{m,n} = \mathcal{R}_{m,n} P_b(\xi)$.

Proof. The polynomial recursion

$$W_m(t) = \frac{t - q^m}{1 - q^m} W_{m-1}(t)$$

becomes (22) after applying it to the shift $n \mapsto n + 1$. $\square$

For $b = 2$, $q = 1/4$, the first filters are

$$\mathcal{R}_{1,n}P = \frac{4P_{n+1} - P_n}{3}, \quad (23)$$

$$\mathcal{R}_{2,n}P = \frac{P_n - 20P_{n+1} + 64P_{n+2}}{45}, \quad (24)$$

$$\mathcal{R}_{3,n}P = \frac{-P_n + 84P_{n+1} - 1344P_{n+2} + 4096P_{n+3}}{2835}. \quad (25)$$

The coefficients are exact rationals; the included program generates arbitrary orders.

4.3 The exact q-moment identity

Lemma 4.4 (all geometric moments). *For every integer $r \geq 0$, let $M_r^{(m)} = \sum_{j=0}^m w_j^{(m)} q^{jr}$. Then*

$$M_r^{(m)} = W_m(q^r). \quad (26)$$

Consequently,

$$M_0^{(m)} = 1, \quad M_r^{(m)} = 0 \quad (1 \leq r \leq m), \quad (27)$$

and for $r \geq m + 1$,

$$M_r^{(m)} = q^{mr} \frac{(q^{1-r}; q)_m}{(q; q)_m} \quad (28)$$

$$= (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q. \quad (29)$$

Proof. Equation (26) follows from the coefficient definition of W_m . The roots give (27). For $r > m$, evaluate the product in (18):

$$W_m(q^r) = \prod_{\ell=1}^m \frac{q^r - q^\ell}{1 - q^\ell} = (-1)^m q^{\binom{m+1}{2}} \prod_{\ell=1}^m \frac{1 - q^{r-\ell}}{1 - q^\ell},$$

which is (29); (28) is the equivalent q-Pochhammer form. $\square$

4.4 The complete error series

Theorem 4.5 (exact q-binomial error). *Let $z = q^n \xi^2$, and suppose $|z| < \pi^2 b^2$. Then*

$$\frac{\mathcal{R}_{m,n}P_b(\xi)}{P_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r=m+1}^{\infty} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) z^r. \quad (30)$$

At a common zero of P_b and all the finite products, the multiplied identity remains valid and both sides vanish. In particular,

$$\frac{\mathcal{R}_{m,n}P_b}{P_b} - 1 \sim (-1)^m q^{\binom{m+1}{2}} \alpha_{m+1}(b) \xi^{2m+2} q^{(m+1)n}. \quad (31)$$

Proof. Insert $\Phi_b(z) = \sum_{r \geq 0} \alpha_r z^r$ into (21). The coefficient of z^r is $\alpha_r W_m(q^r)$. Apply theorem 4.4. $\square$

This identity is substantially stronger than the statement that Richardson extrapolation raises the order. Every residual mode and its exact q-binomial multiplier are known.

4.5 Divided differences and non-asymptotic sign bounds

Let $x_j = q^j z$. The quantity $\sum_j w_j^{(m)} \Phi_b(x_j)$ is the value at zero of the degree- m Lagrange interpolant through the nodes $x_0, \dots, x_m$. The standard interpolation remainder therefore yields a second exact form.

Theorem 4.6 (divided-difference form). *For $0 \leq z < \pi^2 b^2$,*

$$(W_m(T_q)\Phi_b)(z) - 1 = (-1)^m q^{\binom{m+1}{2}} z^{m+1} \Phi_b[0, q^m z, q^{m-1} z, \dots, z], \quad (32)$$

where the bracket denotes a divided difference. Since Φ_b is absolutely monotone on this interval,

$$q^{\binom{m+1}{2}} \alpha_{m+1}(b) z^{m+1} \leq |(W_m(T_q)\Phi_b)(z) - 1| \quad (33)$$

$$\leq \frac{q^{\binom{m+1}{2}} z^{m+1}}{(m+1)!} \Phi_b^{(m+1)}(z). \quad (34)$$

The sign is exactly $(-1)^m$.

Proof. The Lagrange remainder at zero is

$$p_m(0) - \Phi_b(0) = (-1)^m \left(\prod_{j=0}^m x_j \right) \Phi_b[0, x_0, \dots, x_m],$$

and $\prod_j x_j = q^{\binom{m+1}{2}} z^{m+1}$. Because every Taylor coefficient α_r is positive, all derivatives of Φ_b are positive and increasing on the stated interval. The Hermite–Genocchi formula bounds the divided difference between $\Phi_b^{(m+1)}(0)/(m+1)! = \alpha_{m+1}$ and $\Phi_b^{(m+1)}(z)/(m+1)!$. $\square$

Corollary 4.7 (relative bracketing). *For real ξ with $|\xi| < \pi b^{n+1}$ and $P_b(\xi) \neq 0$,*

$$\operatorname{sgn} \left(\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} - 1 \right) = (-1)^m. \quad (35)$$

Thus even and odd extrapolation orders bracket the exact product in relative terms. If $P_b(\xi) > 0$, the bracketing is also an ordinary numerical ordering.

4.6 Uniform stability of arbitrarily high order

Richardson formulae are often numerically unattractive because their coefficient norms grow rapidly. The geometric nodes here behave unusually well.

Theorem 4.8 (exact Lebesgue constant). *The amplification factor of the filter is*

$$\Lambda_m(q) = \sum_{j=0}^m |w_j^{(m)}| = \prod_{r=1}^m \frac{1+q^r}{1-q^r} = \frac{(-q; q)_m}{(q; q)_m}. \quad (36)$$

For fixed $q \in (0, 1)$,

$$\Lambda_\infty(q) = \prod_{r=1}^{\infty} \frac{1+q^r}{1-q^r} < \infty. \quad (37)$$

For $q = 1/4$,

$$\Lambda_\infty(1/4) = 1.969260353668269431947193696967 \dots \quad (38)$$

Proof. The signs in (19) alternate, so

$$\sum_j |w_j^{(m)}| = (-1)^m W_m(-1) = \prod_{r=1}^m \frac{1+q^r}{1-q^r}.$$

The logarithm of the infinite product is $\sum_r [\log(1+q^r) - \log(1-q^r)] = O(\sum_r q^r)$, hence it converges. $\square$

Consequently, absolute input errors bounded by ε produce an output error no larger than $\Lambda_\infty(q)\varepsilon$, uniformly in the extrapolation order. This is a decisive difference from equally spaced high-order extrapolation.

4.7 A rigorous a posteriori certificate

The positive-coefficient error series also compares consecutive truncation levels.

Theorem 4.9 (successive-level certificate). *Fix m and a real ξ . Assume $|\xi| < \pi b^{n+1}$ and $P_b(\xi) \neq 0$. Let $R_n = \mathcal{R}_{m,n}P_b(\xi)$. Then*

$$|R_{n+1} - R_n| \leq |R_n - P_b(\xi)| \leq \frac{|R_{n+1} - R_n|}{1 - q^{m+1}}. \quad (39)$$

Furthermore,

$$P_b(\xi) - R_n \sim \frac{R_{n+1} - R_n}{1 - q^{m+1}}. \quad (40)$$

Proof. By (30), the relative error is $(-1)^m A_n$, where

$$A_n = q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r \xi^{2r} q^{rn} > 0.$$

Every term is multiplied by q^r when n increases by one. Hence $0 < A_{n+1} \leq q^{m+1} A_n$. Therefore

$$(1 - q^{m+1})A_n \leq A_n - A_{n+1} \leq A_n.$$

Multiplying by $|P_b(\xi)|$ proves (39). The leading term has ratio q^{m+1} , which proves (40). $\square$

This certificate is practical: it estimates the unknown infinite product from two consecutive accelerated values and uses only the known number $1 - q^{m+1}$.

5 Physical-space consequences for Rvachev splines

Define the signed extrapolated spline

$$U_{b;m,n}(x) = \sum_{j=0}^m w_j^{(m)} u_{b,n+j}(x). \quad (41)$$

It has total mass one and support in $[-1/(b-1), 1/(b-1)]$. It need not be nonnegative, but

$$\|U_{b;m,n}\|_{L^1} \leq \Lambda_m(q) < \Lambda_\infty(q). \quad (42)$$

Its Fourier transform is $\mathcal{R}_{m,n}P_b$.

5.1 All-order inverse-differential expansion

Theorem 5.1 (all-order spline expansion). *For every pair of nonnegative integers N, s ,*

$$u_{b,n} = \sum_{r=0}^N (-1)^r \alpha_r(b) q^{rn} u_b^{(2r)} + O_{C^s} \left(q^{(N+1)n} \right) \quad (n \rightarrow \infty). \quad (43)$$

Equivalently, as an all-order asymptotic operator identity,

$$u_{b,n} \sim \exp \left(\sum_{k \geq 1} (-1)^k c_k(b) q^{kn} D^{2k} \right) u_b. \quad (44)$$

Proof. Fourier transformation turns $(-1)^r D^{2r} u_b$ into $\xi^{2r} P_b(\xi)$, so the displayed finite sum is exactly the first $N + 1$ terms of (12).

For completeness, split the inverse Fourier integral at $|\xi| = b^n$. On the low-frequency region, $z = q^n \xi^2 \leq 1$, and analyticity of Φ_b gives a uniform Taylor remainder bounded by $C q^{(N+1)n} |\xi|^{2N+2} |P_b(\xi)|$. This remains integrable after multiplication by $|\xi|^s$, because P_b is rapidly decreasing.

On $|\xi| \geq b^n$, the first n factors satisfy

$$|P_{b,n+j}(\xi)| \leq b^{n(n+1)/2} |\xi|^{-n}, \quad j \geq 0,$$

while the remaining factors have modulus at most one. The corresponding weighted Fourier tail is $O(b^{-n^2/2+O(n)})$, hence smaller than every fixed geometric rate. The same bound applies to P_b . Fourier inversion proves the assertion in C^s . $\square$

5.2 Superconvergence of the extrapolated spline

Theorem 5.2 (q-binomial C^∞ expansion). *For fixed m, s and every $N \geq m + 1$,*

$$U_{b;m,n} - u_b = q^{\binom{m+1}{2}} \sum_{r=m+1}^N (-1)^{m+r} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) q^{rn} u_b^{(2r)} + O_{C^s} \left(q^{(N+1)n} \right). \quad (45)$$

In particular,

$$\|U_{b;m,n} - u_b\|_{C^s} \leq C_{b,m,s} q^{(m+1)n}, \quad (46)$$

and the normalized error has the explicit limit

$$q^{-(m+1)n} (U_{b;m,n} - u_b) \longrightarrow -q^{\binom{m+1}{2}} \alpha_{m+1}(b) u_b^{(2m+2)} \quad \text{in } C^s(\mathbb{R}). \quad (47)$$

Proof. Apply inverse Fourier transformation to (30), using $\mathcal{F}^{-1}[\xi^{2r} P_b] = (-1)^r u_b^{(2r)}$. The remainder estimate is the same low/high-frequency argument as in theorem 5.1. At $r = m + 1$, the sign is $(-1)^{m+m+1} = -1$, which gives (47). $\square$

For $b = 2$, each additional extrapolation level changes the basic convergence factor from 4^{-n} to $4^{-(m+1)n}$, while the filter norm remains below 1.970.

5.3 Exact moment matching

The cancellation is stronger than fast convergence: a finite set of moments is already exact.

Theorem 5.3 (exact moment matching). *For every $m, n \geq 0$ and every integer $0 \leq \ell \leq 2m$,*

$$\int_{\mathbb{R}} x^\ell U_{b;m,n}(x) dx = \int_{\mathbb{R}} x^\ell u_b(x) dx. \quad (48)$$

All odd moments vanish by symmetry. The first missed even moment is known exactly:

$$\int_{\mathbb{R}} x^{2m+2} (U_{b;m,n}(x) - u_b(x)) dx = -(2m+2)! q^{\binom{m+1}{2}} \alpha_{m+1}(b) q^{(m+1)n}. \quad (49)$$

Proof. By (30), $\widehat{U}_{b;m,n} - \widehat{u}_b$ has a zero of order at least $2m+2$ at $\xi = 0$. Since

$$\widehat{f}^{(\ell)}(0) = (-i)^\ell \int x^\ell f(x) dx,$$

all moments through degree $2m+1$ agree. Symmetry handles every odd degree. The coefficient of ξ^{2m+2} in the Fourier error is $(-1)^m q^{\binom{m+1}{2}} \alpha_{m+1} q^{(m+1)n}$. Converting the derivative at zero back to the $(2m+2)$ -nd moment contributes $(-1)^{m+1}$, leaving the negative sign in (49). $\square$

Remark 5.4 (quadrature interpretation). For any polynomial p of degree at most $2m$,

$$\int p(x) U_{b;m,n}(x) dx = \int p(x) u_b(x) dx.$$

Thus $U_{b;m,n}$ is a signed, compactly supported, uniformly stable density surrogate with exact polynomial expectations through degree $2m$.

6 Fabius and inverse-Fabius consequences

Under the normalization used throughout the repository, the Fabius function on $[0, 1]$ is the left half of the translated up-function:

$$F(x) = u_2(x-1), \quad 0 \leq x \leq 1. \quad (50)$$

Define the accelerated finite-spline approximation

$$F_{m,n}(x) = U_{2;m,n}(x-1), \quad q = \frac{1}{4}. \quad (51)$$

Corollary 6.1 (accelerated Fabius approximation). *For every $m, s \geq 0$,*

$$\|F_{m,n} - F\|_{C^s([0,1])} = O\left(4^{-(m+1)n}\right), \quad (52)$$

and

$$4^{(m+1)n} (F_{m,n} - F) \longrightarrow -4^{-\binom{m+1}{2}} \alpha_{m+1}(2) F^{(2m+2)} \quad \text{in } C^s([0, 1]). \quad (53)$$

The global statement is possible because all derivatives of the compactly supported up-function vanish at the endpoints. Inversion is more delicate: $F'(x)$ tends to zero at the ends, and this is precisely where the repository's Lambert- W asymptotics become relevant.

Theorem 6.2 (interior inverse-Fabius superconvergence). *Let $K = [a, b] \Subset (0, 1)$, and put $J = F(K)$. Let $G = F^{-1}$ on J . For sufficiently large n , the restriction of $F_{m,n}$ to a neighborhood of K is strictly increasing and has an inverse branch $G_{m,n}$ on J . Then*

$$\|G_{m,n} - G\|_{C^s(J)} = O\left(4^{-(m+1)n}\right) \quad (54)$$

for every fixed s , and uniformly for $y \in J$,

$$4^{(m+1)n}(G_{m,n}(y) - G(y)) \longrightarrow 4^{-\binom{m+1}{2}} \alpha_{m+1}(2) \frac{F^{(2m+2)}(G(y))}{F'(G(y))}. \quad (55)$$

Proof. On a slightly larger compact interval $K' \Subset (0, 1)$, continuity and strict monotonicity give $F' \geq c > 0$. The C^1 convergence in [theorem 6.1](#) implies $F'_{m,n} \geq c/2$ for large n , so the inverse branch exists. Standard differentiation of inverse functions gives (54).

For the leading term, write $F_{m,n} = F + \varepsilon_n$, set $x = G(y)$, and solve

$$0 = F(x + \delta_n) - F(x) + \varepsilon_n(x + \delta_n) = F'(x)\delta_n + \varepsilon_n(x) + o(4^{-(m+1)n}).$$

Equation (53) gives $\varepsilon_n(x) \sim -4^{-\binom{m+1}{2}} \alpha_{m+1} 4^{-(m+1)n} F^{(2m+2)}(x)$, which proves (55). $\square$

6.1 Relation to Lambert– W endpoint asymptotics

The inverse theorem is intentionally restricted to compact interior intervals. Near $y = 0$, the factor $1/F'(G(y))$ becomes large and fixed-order inversion is no longer uniform. The repository’s endpoint theory expresses the dominant scale of $G(y)$ using a Lambert– W variable together with a periodic dyadic phase. The two results therefore occupy complementary limits:

regime	small parameter	governing mechanism
$y \downarrow 0$	$1/\log(1/y)$	Lambert– W and dyadic phase
$n \rightarrow \infty, y \in J \Subset (0, 1)$	4^{-n}	Bernoulli–Bell tail and q-filter.

A genuinely new two-parameter problem is to choose $n = n(y)$ so that (55) remains uniform while $y \downarrow 0$. The exact factor $F^{(2m+2)}(G(y))/F'(G(y))$, combined with the repository’s all-order endpoint expansion, reduces that problem to a concrete matching calculation rather than a vague stability question; see [section 10](#).

7 Numerical verification

All computations in this section used 90-decimal-digit arithmetic. The infinite product was continued until a geometric bound for the omitted logarithmic tail fell below 10^{-70} . The explicit weighted formula and the triangular table (22) were evaluated independently and agreed to the working precision. Symbolic checks verified (29) for a range of orders as an identity in the symbol q .

7.1 Absolute errors

For $b = 2$, the reference values are

$$\begin{aligned} P(1) &= 0.9456037838988178752649922474364995\dots, \\ P(3) &= 0.5857191765350684829125292624510928\dots, \\ P(10) &= -0.03192757801277845828274435480432725\dots \end{aligned}$$

The observed reduction on increasing n is approximately $4^{-(m+1)}$, exactly as predicted.

Table 1: Absolute errors at $\xi = 1$.

n	$m = 0$	$m = 1$	$m = 2$	$m = 3$
3	8.2128×10^{-4}	1.1050×10^{-7}	3.0877×10^{-12}	1.9989×10^{-17}
4	2.0524×10^{-4}	6.9036×10^{-9}	4.8225×10^{-14}	7.8050×10^{-20}
5	5.1304×10^{-5}	4.3143×10^{-10}	7.5345×10^{-16}	3.0485×10^{-22}
6	1.2826×10^{-5}	2.6964×10^{-11}	1.1772×10^{-17}	1.1908×10^{-24}

Table 2: Absolute errors at $\xi = 3$.

n	$m = 0$	$m = 1$	$m = 2$	$m = 3$
3	4.5982×10^{-3}	5.5691×10^{-6}	1.4003×10^{-9}	8.1582×10^{-14}
4	1.1454×10^{-3}	3.4676×10^{-7}	2.1800×10^{-11}	3.1753×10^{-16}
5	2.8608×10^{-4}	2.1652×10^{-8}	3.4031×10^{-13}	1.2392×10^{-18}
6	7.1504×10^{-5}	1.3529×10^{-9}	5.3162×10^{-15}	4.8397×10^{-21}

7.2 Leading constants

At $\xi = 3$, divide the signed error by $q^{(m+1)n}$. The predicted limits from (31) are

m	predicted limit	computed at $n = 11$
0	0.29285958826753424146	0.29285960991271731281
1	-0.022696618090733903713	-0.022696619832602340475
2	0.000365296265765254499	0.000365296293045216575
3	$-1.36214818263598388 \times 10^{-6}$	$-1.36214828256045162 \times 10^{-6}$

The alternating signs and limiting constants are resolved without fitting any free parameter.

7.3 A posteriori estimates

For example, at $\xi = 3, n = 4, m = 3$, the true absolute error is $3.17531383672 \times 10^{-16}$. The computable difference between successive accelerated values is $3.16292144780 \times 10^{-16}$, and [theorem 4.9](#) gives the upper bound $3.17532506132 \times 10^{-16}$. The certificate therefore encloses the true error in an interval of relative width about 4×10^{-6} , using no reference value of $P(3)$.

8 A universal geometric-product theorem

The q-filter is not tied to the sine function. Its universality clarifies which parts of the construction are special to Rvachev's function.

Theorem 8.1 (universal analytic-factor acceleration). *Let φ be an even analytic function near zero with $\varphi(0) = 1$, nonzero in a neighborhood of zero, and*

$$\log \varphi(w) = - \sum_{k=1}^{\infty} a_k w^{2k}. \quad (56)$$

For $b > 1$, define

$$Q_b(\xi) = \prod_{\nu \geq 1} \varphi(\xi/b^\nu), \quad Q_{b,n}(\xi) = \prod_{\nu=1}^n \varphi(\xi/b^\nu).$$

Table 3: Absolute errors at $\xi = 10$, beyond the first sign change of the infinite product but still inside the tail disk for the displayed n .

n	$m = 0$	$m = 1$	$m = 2$	$m = 3$
3	2.9276×10^{-3}	3.9468×10^{-5}	1.1012×10^{-7}	7.1217×10^{-11}
4	7.0230×10^{-4}	2.3635×10^{-6}	1.6504×10^{-9}	2.6705×10^{-13}
5	1.7380×10^{-4}	1.4617×10^{-7}	2.5525×10^{-11}	1.0327×10^{-15}
6	4.3341×10^{-5}	9.1120×10^{-9}	3.9782×10^{-13}	4.0240×10^{-18}

Then locally at $q^n \xi^2 = 0$,

$$\frac{Q_{b,n}(\xi)}{Q_b(\xi)} = \exp \left(\sum_{k \geq 1} \frac{a_k}{b^{2k} - 1} \xi^{2k} q^{kn} \right). \quad (57)$$

Let its Taylor coefficients in $z = q^n \xi^2$ be β_r . The same weights (19) satisfy

$$\frac{\sum_{j=0}^m w_j^{(m)} Q_{b,n+j}(\xi)}{Q_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \beta_r z^r. \quad (58)$$

If all $a_k \geq 0$, then all $\beta_r \geq 0$, and the sign law, divided-difference bounds, and a posteriori certificate remain valid.

Proof. The proof of [theorem 3.3](#) uses only the geometric sum $\sum_{\ell \geq 1} b^{-2k\ell} = 1/(b^{2k} - 1)$. The q-filter calculation then uses only the Taylor expansion and [theorem 4.4](#). $\square$

Thus the sinc-specific information is concentrated in the Bernoulli values of $a_k = \zeta(2k)/(k\pi^{2k})$. The q-binomial extrapolation, its stability, and its exact mode cancellation belong to the geometry of the scales rather than to the elementary factor.

9 Lean formalization roadmap

The analytic proof naturally separates into formalization-sized modules.

1. **Finite q-algebra.** Define $(q; q)_n$, Gaussian binomial coefficients, W_m , and the weights. Prove $W_m(q^r) = 0$ for $1 \leq r \leq m$ and (29). This part is purely algebraic.
2. **Formal power series.** Work first in a characteristic-zero commutative ring. From $\Phi = \exp(\sum c_k z^k)$, derive the Bell recurrence (15); then prove the coefficientwise filtered identity (30). This avoids analytic convergence in the first pass.
3. **Analytic sinc logarithm.** Establish (8) in a disk, sum the geometric scales, and identify the analytic power series with the actual product. Existing repository lemmas about the sinc product should reduce the amount of new infrastructure.
4. **Interpolation and positivity.** Formalize the Lagrange weights, the divided-difference remainder, and absolute monotonicity of a power series with nonnegative coefficients. The sign theorem can also be proved directly from the positive coefficient series if divided differences are inconvenient.
5. **Fourier/measure layer.** Define the uniform box measures and their finite convolutions. Moment matching follows from derivatives of the characteristic function at zero and may be formalized before the full C^∞ theorem.

6. **Asymptotic layer.** Prove low-frequency Taylor bounds and the high-frequency superexponential estimate. This is the technically heaviest step, but it is independent of q -binomial algebra.
7. **Inverse-function layer.** Apply a quantitative inverse function theorem on compact intervals where F' has a positive lower bound, then derive (55) from the first-order perturbation identity.

The accompanying Python file supplies exact test vectors for the algebraic modules and high-precision regression tests for the analytic statements.

10 Further frontier problems

10.1 A matched Lambert–Richardson endpoint theory

Let $G = F^{-1}$ and let $G_{m,n}$ be the accelerated inverse branch. Determine the largest endpoint region $0 < y \leq y_n$ on which

$$G_{m,n}(y) - G(y) \sim 4^{-\binom{m+1}{2}} \alpha_{m+1}(2) 4^{-(m+1)n} \frac{F^{(2m+2)}(G(y))}{F'(G(y))}$$

remains uniform. Substitution of the repository’s Lambert– W expansion for $G(y)$ should identify a sharp transition curve y_n , together with a dyadically periodic modulation. This would connect two asymptotic theories that are presently separate.

10.2 Optimal order in finite precision

Although $\Lambda_m(1/4) < 1.970$, the finite products themselves may suffer relative loss near zeros. A complete finite-precision theory should choose m and n adaptively from: (i) the certificate (39); (ii) the distance to the shared zero set; and (iii) the working precision. Because the filter norm is uniformly bounded, one expects the optimal order to be governed mainly by product evaluation rather than extrapolation instability.

10.3 Positivity restoration

The signed density $U_{b;m,n}$ has exact low moments but need not be positive. Is there a nonlinear, positivity-preserving correction that retains $q^{(m+1)n}$ convergence and the moment constraints? One possibility is a small entropy projection onto the convex set of nonnegative densities with the same moments. The exact first missed moment (49) provides a natural scale for such a correction.

10.4 Dyadic rational evaluation from accelerated splines

At a dyadic point, each finite spline $u_{2,n+j}$ is exactly computable from integrated Thue–Morse data and q -binomial formulae already present in the repository. Substituting those values into (20) should yield explicit rational or q -rational approximants to $F(x)$ with a certified $4^{-(m+1)n}$ remainder. This would produce a new bridge between exact dyadic arithmetic and the analytic sinc tail.

10.5 Complex domains and pole-aware extrapolation

The nearest tail pole at $z = \pi^2 b^2$ controls the Taylor expansion. For complex ξ , one can conformally map the slit z -plane before applying a geometric interpolation filter. A pole-aware variant may retain exact q -mode cancellation while extending useful acceleration beyond the first omitted sinc zero.

10.6 Other refinable and compactly supported functions

Apply [theorem 8.1](#) to elementary factors arising from compactly supported refinable distributions, Bernoulli convolutions, or symmetric B-splines. The main classification question is: which elementary factors have nonnegative logarithmic cumulants, and therefore inherit the exact sign law and successive-level certificate?

11 Conclusion

The truncation of the Rvachev sinc product carries more structure than a generic convergent product. Its omitted tail is an analytic function of the single geometric variable $z = 4^{-n}\xi^2$. Bernoulli numbers describe the logarithm of that function, Bell polynomials describe the function itself, and Lagrange interpolation on the nodes $1, q, \dots, q^m$ produces a q -binomial Richardson filter with a complete, sign-definite error series.

The most notable consequences are exact rather than heuristic: the q -moments of the weights are known in closed form; the filter norm stays bounded at arbitrarily high order; consecutive accelerated levels certify the error; and the associated signed box splines match the first $2m$ moments of the limiting up-density for every n . Fourier inversion converts the same algebra into C^∞ -superconvergence with an explicit derivative profile. Translation and local inversion then give a new interior approximation theory for the Fabius function and its inverse, complementary to the repository's Lambert- W endpoint asymptotics.

These results create concrete next steps: a Lean formalization beginning with the finite q -algebra, a dyadic exact-evaluation implementation, and a matched two-parameter theory linking truncation acceleration to the periodic Lambert phase near the endpoint.

A Exact coefficients and filters at $b = 2$

The next two Bell coefficients, useful for regression tests, are

$$\alpha_6 = \frac{1884491755011469}{1980124051433319792000000},$$

$$\alpha_7 = \frac{151616054172597829}{6278781742186853835936000000}.$$

The order-four filter is

$$\mathcal{R}_{4,n}P = \frac{1}{722925}P_n - \frac{4}{8505}P_{n+1} + \frac{64}{2025}P_{n+2} - \frac{4096}{8505}P_{n+3} + \frac{1048576}{722925}P_{n+4}.$$

The order-five weights are

$$\left(-\frac{1}{739552275}, \frac{4}{2168775}, -\frac{64}{127575}, \frac{4096}{127575}, -\frac{1048576}{2168775}, \frac{1073741824}{739552275} \right).$$

Despite the large integer numerators, the sum of absolute values is below 1.968 at order five.

B Corpus audit map

This appendix records the source families used to establish the non-duplication boundary. Historical variants were read for content rather than treated as independent discoveries.

Current top-level expositions

- `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`
- `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`
- `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex`
- `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`
- `non-formalized-research-frontiers/non-formalized-research-frontiers.tex`

Current frontier drafts

- `drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex`
- LaTeX source in `Rvachev_Up_Fourier_Decay-2`
- LaTeX source in `Rvachev_Up_Fourier_Decay-3`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Comparative_Audit`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Gentle_Guide`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Second_Wave_Audit`
- LaTeX source in `asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions`
- LaTeX source in `rvachev_up_fourier_decay-1`
- LaTeX sources in `rvachev_up_fourier_decay-4` through `rvachev_up_fourier_decay-8`

Paper-source directories

- `AIF_2017__67_2_637_0.tex` and its corrected source
- `Lacunary_products.tex` (arXiv:1502.06738v2)
- `09-Function.tex` (arXiv:1702.05442v1)
- `157-Arithmetic-v3.tex` (arXiv:1702.06487v3)
- `document.tex` (arXiv:2211.13570v1)
- `ubiq15-corrected.tex`

Historical archive families

- `Fabius_Asymptotic.tex`
- `K-fold summation over signed Thue-Morse.tex`
- `Non_Elementarity_of_the_Fabius_Function.tex`
- `Small_Argument_Asymptotics.tex`
- `fabius-inverse-dyadic-closed-form.tex`
- Historical primary exposition variants under `Fabius_Function_and_Rvachev_Up`, `Fabius_and_Rvachev_Up-2`, `Fabius_and_Rvachev_up`, and `Fabius_function_and_up_function`
- `Fabius_Function_Missing_Parts` variants

- `Fabius_q_Special_Functions` and `fabius-q-special-functions`
- `Fabius_Function_Lean_Walkthrough` and `fabius_lean_walkthrough` variants
- `FabiusFunction_Mathematical_Glossary` variants
- `Fabius_Dyadic_Asymptotic_Bridge`
- `Fabius_Dyadic_Formulae_and_Alternative_Representations`
- `Fabius_Dyadic_Formulae_to_Asymptotics`
- `Fabius_Dyadic_q_Connections`
- `Fabius_Thue_Morse_Convergence_Rate`
- `Fabius_Thue_Morse_Convergence_Rates`
- `Primary_Exposition_Gap_Register`
- `Repeated_Integration_and_Rvachev_Up`
- `Rvachev_Up_from_Repeated_Integration`
- Earlier consolidated `non-formalized-research-frontiers` source
- `Thue_Morse_Formula_Atlas-1`, `Thue_Morse_Formula_Atlas-2`, and `Thue_Morse_Formula_Atlas-3`

The current consolidated frontier source explicitly records the absorption of several former notebooks and their checksums. That provenance was used when checking whether a proposed statement was genuinely absent or merely moved.

C Reproducibility notes

The archive accompanying this report contains:

- `q_richardson_sinc_frontier.tex` — the present source;
- `q_richardson_sinc_frontier.pdf` — the rendered report;
- `verify_frontier_results.py` — exact symbolic and high-precision numerical checks;
- `numerical_results.txt` and `numerical_results.csv` — machine-readable output;
- `README.txt` — compilation and reproduction commands.

The script checks the q -moment identity symbolically for several arbitrary orders, generates the exact Bernoulli and Bell coefficients, verifies agreement between the explicit and triangular filters, reproduces tables 1 to 3, checks the leading constants, and asserts the two-sided certificate (39) at representative points.

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